Tema 4: Interfaces Gráficas con Swing
Parte II - Layouts

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(Basado en el material creado por Puri Arenas, Jesús Correas y Samir Genaim)

Colocando Componentes en un JFrame

- Las componentes en un frame se añaden utilizando su **content pane**
- Hasta ahora hemos utilizado un layout vacío (**null**), y hemos colocado las componentes indicando su tamaño y su localización. Qué inconvenientes tiene esto?
  1. Las componentes se colocan de forma manual
  2. Hay que darles posición y tamaño
  3. Cuando se redimensiona la ventana las componentes permanencen ancladas, no se mueven
  4. El cambio de SO, resolución, pantalla o fuente de letra puede afectar a la visualización
- En Java se pueden fijar distintos **layouts**, que dividen la ventana en zonas en las que colocar componentes de forma simple, sin necesidad de especificar localización y tamaño. A través del método **pack()**, el tamaño de la ventana se ajustará al tamaño de las componentes. **Si se fija un layout distinto de null, entonces no se pueden colocar las componentes de forma manual**

Colocando Componentes en un JFrame

```
JPanel mainPanel = new JPanel();
mainPanel.setLayout(null );
JPanel redPanel = new JPanel();
redPanel. setBackground(Color.red);
redPanel. setLocation (10,  10);
redPanel. setSize (50,50);
mainPanel.add(redPanel);
```

Layout Managers en Swing

- Los layouts disponibles en Swing son
  1. Flow Layout (JPanel lo usa por defecto)
  2. Border Layout (JFrame lo usa por defecto)
  3. Box Layout
  4. Grid Layout
  5. Grid Bag Layout
  6. Group Layout
  7. Spring Layout

FlowLayout

- **Coloca las componentes en fila**, en función del tamaño de la ventana
- Se puede especificar desde donde empezar a colocar las componentes en caso de redimensionar la ventana: de izquierda a derecha, centradas, o de derecha a izquierda.**Hay que tener en cuenta que si las componentes caben en una fila, esa será su disposición.**
- Es conveniente utilizar el método **pack()** del frame para ajustar el tamaño de la ventana con respecto a las componentes.
- La constructora más habitual para crear un **FlowLayout** es:

  new FlowLayout(FlowLayout.CONSTANT, int vspace, int hspace)

  - **CONSTANT** puede ser **LEFT, RIGHT, CENTER**
  - **hspace** nos da la separación horizontal entre las componentes
  - **vspace** nos da la separación entre las componentes cuando se hace resize

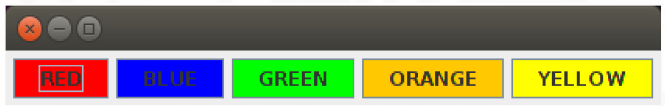
FlowLayout

```
private void initGUI() {
  FlowLayout layout = new FlowLayout(FlowLayout.LEFT,5,5);
  Container contentPane = this.getContentPane();
  contentPane.setLayout(layout);

  contentPane.add(createButton(Color.red,"RED"));
  contentPane.add(createButton(Color.blue,"BLUE"));
  contentPane.add(createButton(Color.green,"GREEN"));
  contentPane.add(createButton(Color.orange,"ORANGE"));
  contentPane.add(createButton(Color.yellow,"YELLOW"));

  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();   // ajusta la ventana al tamaño de las componentes
  this.setVisible(true);
}

private JButton createButton(Color color, String caption) {
  JButton button = new JButton(caption);
  button.setBackground(color);
  return button;
}
```



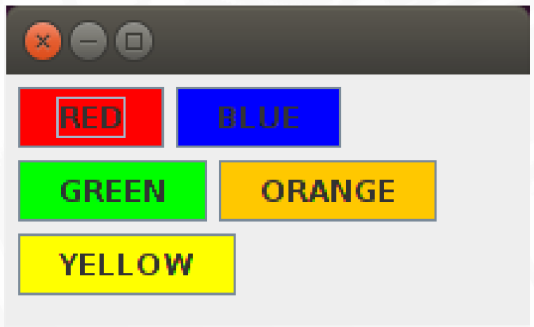
FlowLayout

```
private void initGUI() {
  FlowLayout layout = new FlowLayout(FlowLayout.LEFT,5,5);
  Container contentPane = this.getContentPane();
  contentPane.setLayout(layout);

  contentPane.add(createButton(Color.red,"RED"));
  contentPane.add(createButton(Color.blue,"BLUE"));
  contentPane.add(createButton(Color.green,"GREEN"));
  contentPane.add(createButton(Color.orange,"ORANGE"));
  contentPane.add(createButton(Color.yellow,"YELLOW"));

  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();   // ajusta la ventana al tamaño de las componentes
  this.setVisible(true);
}

private JButton createButton(Color color, String caption) {
  JButton button = new JButton(caption);
  button.setBackground(color);
  return button;
}
```



Al hacer resize las componentes se colocan de izquierda a derecha

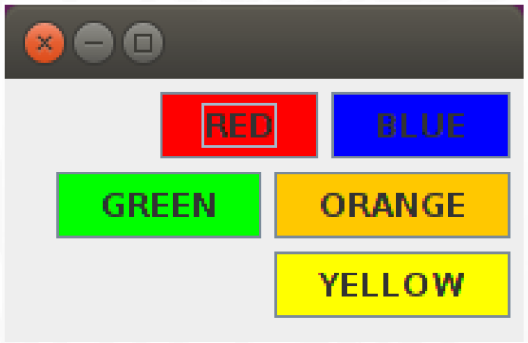
FlowLayout

```
private void initGUI() {
  FlowLayout layout = new FlowLayout(FlowLayout.RIGHT,5,5);
  Container contentPane = this.getContentPane();
  contentPane.setLayout(layout);

  contentPane.add(createButton(Color.red,"RED"));
  contentPane.add(createButton(Color.blue,"BLUE"));
  contentPane.add(createButton(Color.green,"GREEN"));
  contentPane.add(createButton(Color.orange,"ORANGE"));
  contentPane.add(createButton(Color.yellow,"YELLOW"));

  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();   // ajusta la ventana al tamaño de las componentes
  this.setVisible(true);
}

private JButton createButton(Color color, String caption) {
  JButton button = new JButton(caption);
  button.setBackground(color);
  return button;
}
```



De derecha a izquierda

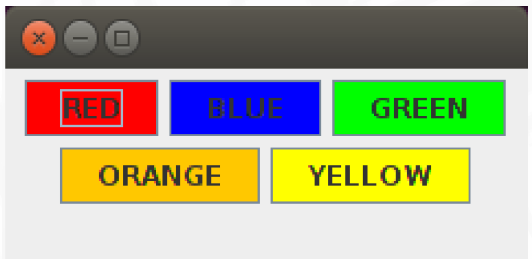
FlowLayout

```
private void initGUI() {
  FlowLayout layout = new FlowLayout(FlowLayout.CENTER,5,5);
  Container contentPane = this.getContentPane();
  contentPane.setLayout(layout);

  contentPane.add(createButton(Color.red,"RED"));
  contentPane.add(createButton(Color.blue,"BLUE"));
  contentPane.add(createButton(Color.green,"GREEN"));
  contentPane.add(createButton(Color.orange,"ORANGE"));
  contentPane.add(createButton(Color.yellow,"YELLOW"));

  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();   // ajusta la ventana al tamaño de las componentes
  this.setVisible(true);
}

private JButton createButton(Color color, String caption) {
  JButton button = new JButton(caption);
  button.setBackground(color);
  return button;
}
```

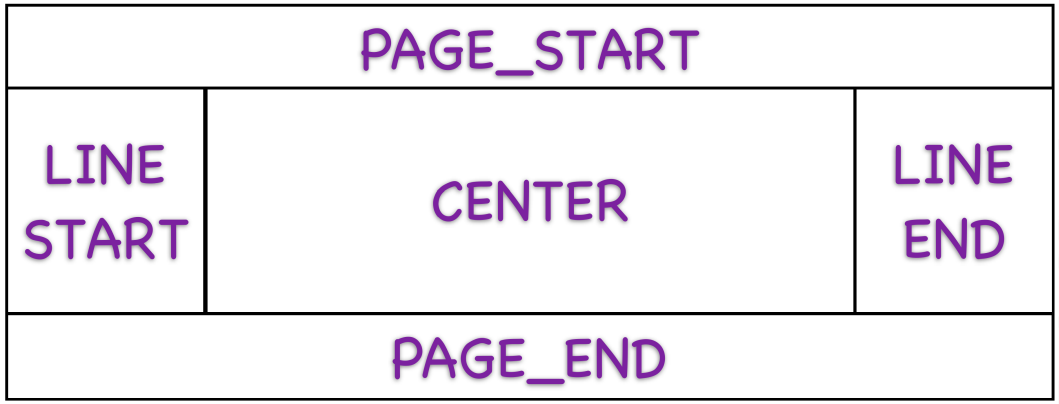


Centro

BorderLayout

Es el layout por defecto para JFrame y JDialog.

- Divide la ventana en 5 partes: centro, arriba, abajo, derecha e izquierda.

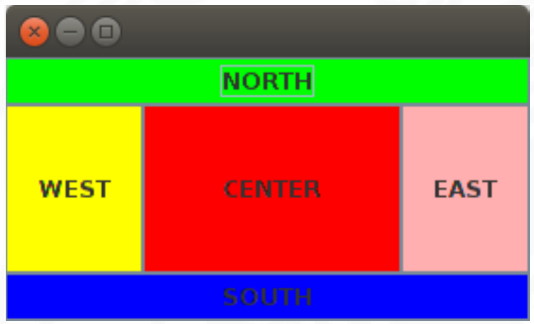


  Es adecuado para diseñar ventanas en las que hay una componente central importante (tabla, lista, etc.) y tiene menús o barras de herramientas situados arriba, abajo, a la derecha o a la izquierda.
  Es preferible usar PAGE_START, PAGE_END, LINE_START y LINE_END ya que son independientes del sistema de coordenadas del SO.

- Al redimensionar la parte superior e inferior no cambian de altura pero si de anchura. Izquierda y derecha no cambian de anchura pero si de altura. El centro cambia en altura y en anchura. Las componentes si se redimensionan proporcionalmente.

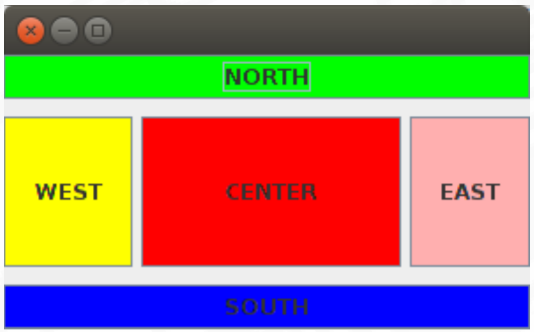
BorderLayout

```
private void initGUI() {
  this.setDefaultCLoseOperation(JFrame.EXIT_ON_CLOSE);
  this.setLayout(new BorderLayout());
  this.add (createBUtton(Color.RED, ''CENTER''), BorderLayout.CENTER);
  this.add (createButton(Color.GREEN,"NORTH"), BorderLayout.NORTH);
  this.add (createButton(Color.BLUE,"SOUTH"), BorderLayout.SOUTH);
  this.add (createButton(Color.YELLOW,"WEST"), BorderLayout.WEST);
  this.add (createButton(Color.PINK,"EAST"), BorderLayout.EAST);
  this.setVisible (true);
  this.pack();
  }
```



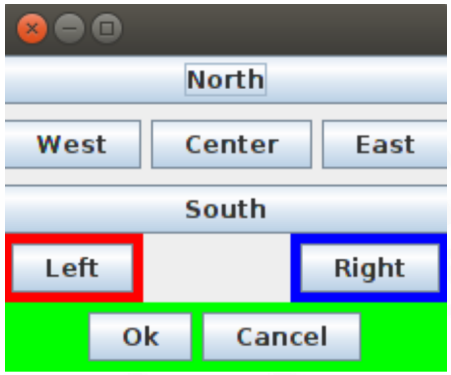
BorderLayout

```
private void initGUI() {
  this.setDefaultCLoseOperation(JFrame.EXIT_ON_CLOSE);
  this.setLayout(new BorderLayout(5,10)); //separacion vertical y horizontal
  this.add (createBUtton(Color.RED, "CENTER"), BorderLayout.CENTER);
  this.add (createButton(Color.GREEN,"NORTH"), BorderLayout.NORTH);
  this.add (createButton(Color.BLUE,"SOUTH"), BorderLayout.SOUTH);
  this.add (createButton(Color.YELLOW,"WEST"), BorderLayout.WEST);
  this.add (createButton(Color.PINK,"EAST"), BorderLayout.EAST);
  this.setVisible (true);
  this.pack();
  }
```

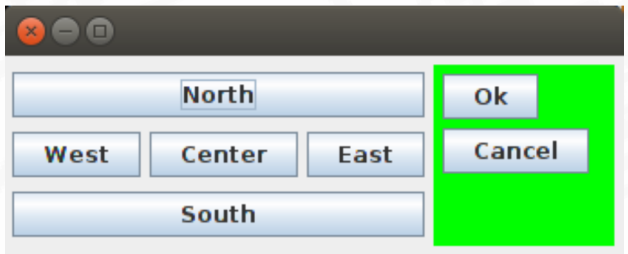


Layouts y Componentes

- Un **layout** en Java **solo permite añãdir una componente en una zona de la ventana**. En un **BorderLayout** solo podríamos colocar, por ejemplo, un botón en cada zona de la pantalla. Pero no podríamos tener:

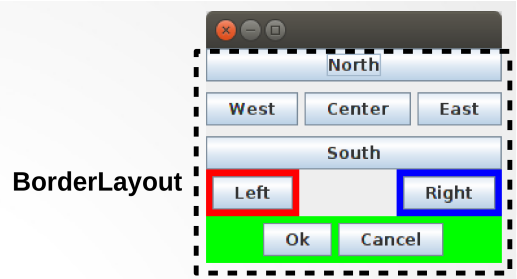


- Tampoco podríamos construir:



- La solución es utilizar **paneles (JPanel), que son contenedores pero también componentes. Al ser un contenedor puede tener una componente, y en concreto al ser el panel una componente además se pueden anidar. Es decir, podemos tener paneles dentro de paneles.**

Layouts y Componentes



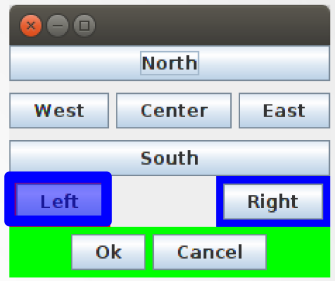

```
private void initGUI() {
  this.setDefaultCLoseOperation(JFrame.EXIT_ON_CLOSE);
  JPanel miPanel = new JPanel(new BorderLayout()); //crea un panel con un BorderLayout

  mainPanel.add(createLateralPanel(Color.RED,"Left"),BorderLayout.WEST);
  mainPanel.add(createLateralPanel(Color.BLUE,"Right"),BorderLayout.EAST);
  mainPanel.add(createSuperiorPanel(), BorderLayout.NORTH);
  mainPanel.add(createInferiorPanel (), BorderLayout.SOUTH);

  this.add(miPanel); //añade el panel a la ventana principal
  this.setVisible (true);
  this.pack();
}
```

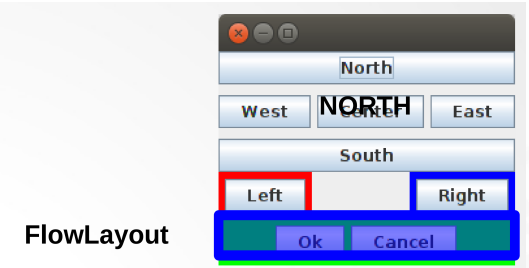
Layouts y Componentes

FlowLayout




```
private JPanel createLateralPanel (Color color, String caption){
  JPanel sidePanel = new JPanel();
  sidePanel.setLayout(new FlowLayout());
  sidePanel.setBackground(color); //especifica el color del panel
  sidePanel.add(this.createButton(caption));
  return sidePanel;
}
```

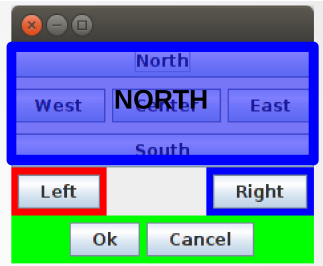
Layouts y Componentes




```
private JPanel createInferiorPanel () {
  JPanel infPanel = new JPanel(new FlowLayout());
  infPanel.setBackground(Color.green); // color de fondo del panel
  infPanel.add(createButton("Ok"));
  infPanel.add(createButton("Cancel"));
  return infPanel;
}
```

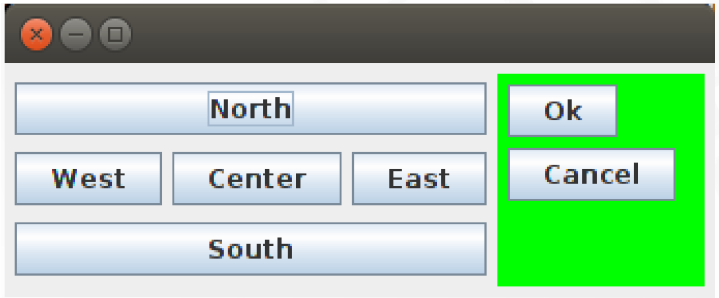
Layouts y Componentes

BorderLayout

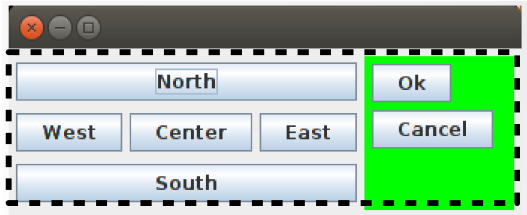



```
private JPanel createSuperiorPanel () {
  JPanel supPanel = new JPanel();
  supPanel.setLayout(new BorderLayout(5,8));
  supPanel.add(createButton("North"),BorderLayout.NORTH);
  supPanel.add(createButton("South"),BorderLayout.SOUTH);
  supPanel.add(createButton("East"),BorderLayout.EAST);
  supPanel.add(createButton("West"),BorderLayout.WEST);
  supPanel.add(createButton("Center"),BorderLayout.CENTER);
  return supPanel;
}
```

Layouts y Componentes



Layouts y Componentes



FlowLayout

```
private void initGUI() {
  JPanel mainPanel = new JPanel(new FlowLayout());


  this.add(mainPanel);
  this.setVisible (true);
  this.pack();
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}
```

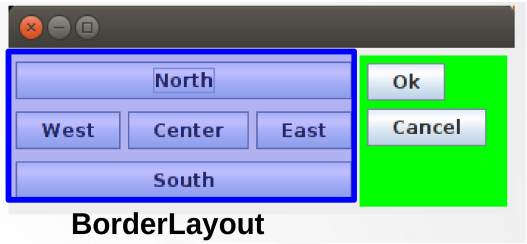
Layouts y Componentes

```
private void initGUI() {
  JPanel mainPanel = new JPanel(new FlowLayout());

  mainPanel.add(createLeftPanel())
  mainPanel.add(createRightPanel());

  this.add(mainPanel);
  this.setVisible (true);
  this.pack();
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}

private JPanel createLeftPanel() {
  JPanel leftPanel = new JPanel();
  leftPanel.setLayout(new BorderLayout(5,8));
  leftPanel.add(createButton("North"),BorderLayout.NORTH);
  leftPanel.add(createButton("South"),BorderLayout.SOUTH);
  leftPanel.add(createButton("East"),BorderLayout.EAST);
  leftPanel.add(createButton("West"),BorderLayout.WEST);
  leftPanel.add(createButton("Center"),BorderLayout.CENTER);
  return leftPanel;
}
```



BorderLayout

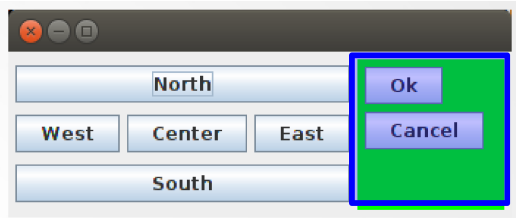
Layouts y Componentes

```
private void initGUI() {
  JPanel mainPanel = new JPanel(new FlowLayout());

  mainPanel.add(createLeftPanel())
  mainPanel.add(createRightPanel());

  this.add(mainPanel);
  this.setVisible (true);
  this.pack();
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
}

private JPanel createRightPanel() {
  // los botones sino caben se alinean desde la izquierda
  JPanel rightPanel = new JPanel(new FlowLayout(FlowLayout.LEFT));
  // se fija el tamaño del panel
  rightPanel.setPreferredSize (new Dimension(100,100));
  rightPanel.setBackground(Color.green);
  rightPanel.add(createButton("Ok"));
  rightPanel.add(createButton("Cancel"));
  return rightPanel;
}
}
```



FlowLayout

BoxLayout

- Se comporta como **FlowLayout**, pero permite colocar los elementos en horizontal o en vertical
- En horizontal:
  frame.setLayout(new BoxLayout(contentPane, BoxLayout.X_AXIS));
  frame.setLayout(new BoxLayout(contentPane, BoxLayout.LINE_AXIS));
- En vertical:
  frame.setLayout(new BoxLayout(contentPane, BoxLayout.Y_AXIS));
  frame.setLayout(new BoxLayout(contentPane, BoxLayout.PAGE_AXIS));
- Es preferible usar la segunda opción, que es independiente del sistema de coordenada usado por el sistema operativo.
- No se redimensionan las componentes cuando se redimensiona la ventana.

BoxLayout

```
private void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.Y_AXIS));
  addButton("Holaa", mainPanel);
  addButton("Holaaa", mainPanel);
  addButton("Holaaaa", mainPanel);
  addButton("Holaaaaa", mainPanel);
  addButton("Holaaaaaa", mainPanel);
  this.setContentPane(mainPanel); // this.add(mainPanel);
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();
  this.setVisible (true);
}

private void addButton(String text, JPanel container) {
  JButton button = new JButton(text);
  button.setAlignmentX(JButton.CENTER_ALIGNMENT);

  // fija el tamaño de la componente (500 de ancho por 100 de largo)
  button.setPreferredSize (new Dimension(500, 100));
  // permite ampliar proporcionalmente hasta 1000 de ancho por 150 de alto.
  button.setMaximumSize(new Dimension(1000, 150));
  // permite disminuir proporcionalmente hasta 200 de ancho por 30 de alto
  button.setMinimumSize(new Dimension(200,30));
  contained.add(button);
}
```

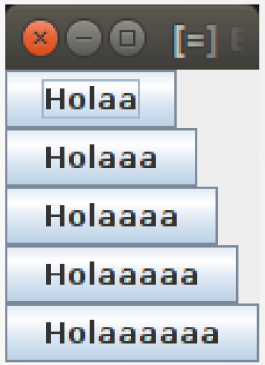


BoxLayout

```
private void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.Y_AXIS));
  addButton("Holaa", mainPanel);
  addButton("Holaaa", mainPanel);
  addButton("Holaaaa", mainPanel);
  addButton("Holaaaaa", mainPanel);
  addButton("Holaaaaaa", mainPanel);
  this.setContentPane(mainPanel); // this.add(mainPanel);
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();
  this.setVisible (true);
}

private void addButton(String text, JPanel container) {
  JButton button = new JButton(text);
  button.setAlignmentX(JButton.LEFT_ALIGNMENT);

  // fija el tamaño de la componente (500 de ancho por 100 de largo)
  button.setPreferredSize (new Dimension(500, 100));
  // permite ampliar proporcionalmente hasta 1000 de ancho por 150 de alto.
  button.setMaximumSize(new Dimension(1000, 150));
  // permite disminuir proporcionalmente hasta 200 de ancho por 30 de alto
  button.setMinimumSize(new Dimension(200,30));
  contained.add(button);
}
```

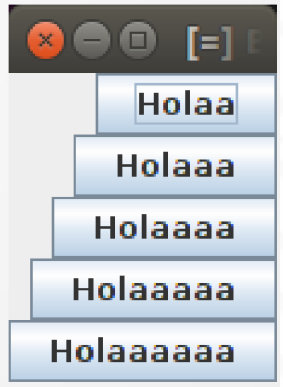


BoxLayout

```
private void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.Y_AXIS));
  addButton("Holaa", mainPanel);
  addButton("Holaaa", mainPanel);
  addButton("Holaaaa", mainPanel);
  addButton("Holaaaaa", mainPanel);
  addButton("Holaaaaaa", mainPanel);
  this.setContentPane(mainPanel); // this.add(mainPanel);
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.pack();
  this.setVisible (true);
}

private void addButton(String text, JPanel container) {
  JButton button = new JButton(text);
  button.setAlignmentX(JButton.RIGHT_ALIGNMENT);

  // fija el tamaño de la componente (500 de ancho por 100 de largo)
  button.setPreferredSize (new Dimension(500, 100));
  // permite ampliar proporcionalmente hasta 1000 de ancho por 150 de alto.
  button.setMaximumSize(new Dimension(1000, 150));
  // permite disminuir proporcionalmente hasta 200 de ancho por 30 de alto
  button.setMinimumSize(new Dimension(200,30));
  contained.add(button);
}
```



BoxLayout




```
button.setPreferredSize (new Dimension(500, 100));
```

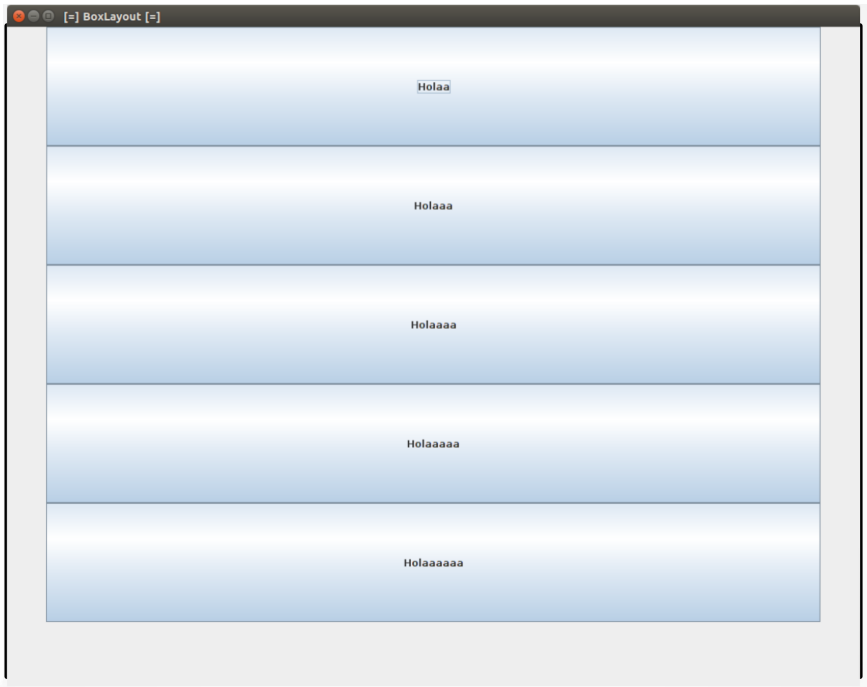
BoxLayout




```
button.setMaximumSize(new Dimension(1000, 150));
```

Al ensanchar la ventana los botones crecen proporcionalmente hasta tamaño 1000

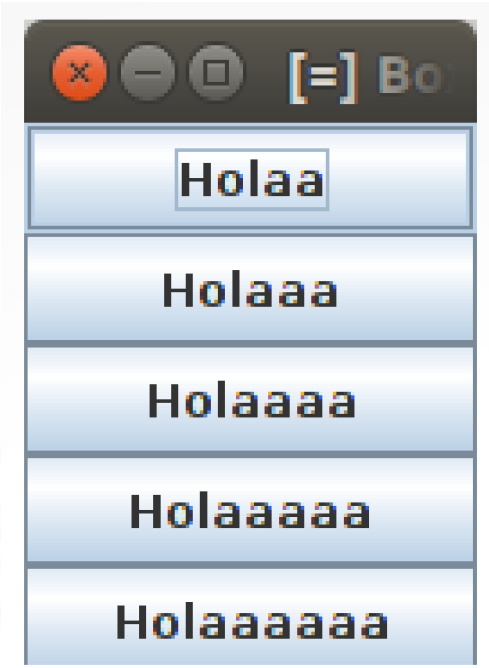
BoxLayout



button.setMaximumSize(new Dimension(1000, 150));

Al alargar la ventana los botones crecen proporcionalmente hasta un largo de 150

BoxLayout




```
button.setMinimumSize(new Dimension(200, 30));
```

Al encoger la ventana los botones decrecen proporcionalmente hasta un largo de 30 y un ancho de 200

BoxLayout

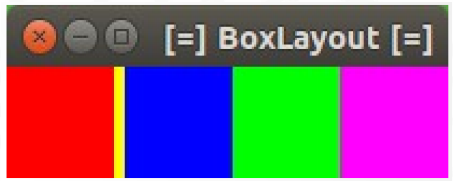



```
mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
```

BoxLayout

```
public void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
  mainPanel.add(createPanel(Color.red, 50, 50));
  mainPanel.add(createPanel(Color.yellow, 5, 50));
  mainPanel.add(createPanel(Color.blue, 50, 10));
  mainPanel.add(createPanel(Color.green, 50, 20));
  mainPanel.add(createPanel(Color.magenta, 50, 50));
  ...
}

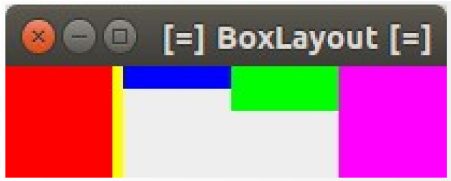
private JPanel createPanel(Color color, int x, int y) {
  JPanel panel = new JPanel();
  panel.setBackground(color);
  panel.setAlignmentY(JPanel.TOP_ALIGNMENT);
  panel.setPreferredSize (new Dimension(x, y));
  return panel;
}
```



BoxLayout

```
public void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
  mainPanel.add(createPanel(Color.red, 50, 50));
  mainPanel.add(createPanel(Color.yellow, 5, 50));
  mainPanel.add(createPanel(Color.blue, 50, 10));
  mainPanel.add(createPanel(Color.green, 50, 20));
  mainPanel.add(createPanel(Color.magenta, 50, 50));
  ...
}

private JPanel createPanel(Color color, int x, int y) {
  JPanel panel = new JPanel();
  panel.setBackground(color);
  panel.setAlignmentY(JPanel.TOP_ALIGNMENT);
  panel.setPreferredSize (new Dimension(x, y));
  panel.setMaximumSize(new Dimension(x, y));
  panel.setMinimumSize(new Dimension(x, y));
  return panel;
}
```



Si fijamos el tamaño máximo y el mínimo, entonces si salen con su tamaño real

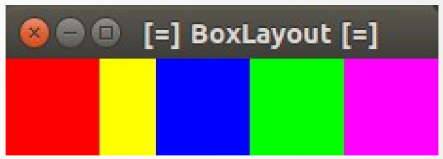
BoxLayout-Área rígida

```
public void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
  mainPanel.add(createPanel(Color.red, 50, 50));

  mainPanel.add(createPanel(Color.yellow, 5, 50));
  mainPanel.add(createPanel(Color.blue, 50, 10));

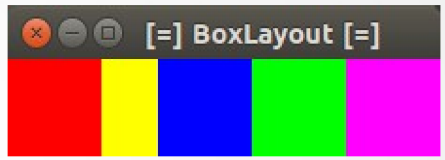
  mainPanel.add(createPanel(Color.green, 50, 20));
  mainPanel.add(createPanel(Color.magenta, 50, 50));
  ...
}

private JPanel createPanel(Color color, int x, int y) {
  JPanel panel = new JPanel();
  panel.setBackground(color);
  panel.setAlignmentY(JPanel.TOP_ALIGNMENT);
  panel.setPreferredSize (new Dimension(x, y));
  return panel;
}
```

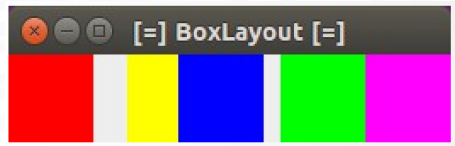


BoxLayout-Área rígida

```
public void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
  mainPanel.add(createPanel(Color.red, 50, 50));
  mainPanel.add(Box.createRigidArea(new Dimension(20, 0)));
  mainPanel.add(createPanel(Color.yellow, 5, 50));
  mainPanel.add(createPanel(Color.blue, 50, 10));
  mainPanel.add(Box.createRigidArea(new Dimension(10, 0)));
  mainPanel.add(createPanel(Color.green, 50, 20));
  mainPanel.add(createPanel(Color.magenta, 50, 50));
  ...
}
```




```
private JPanel createPanel(Color color, int x, int y) {
  JPanel panel = new JPanel();
  panel.setBackground(color);
  panel.setAlignmentY(JPanel.TOP_ALIGNMENT);
  panel.setPreferredSize (new Dimension(x, y));
  return panel;
}
```



Box.createRigidArea(new Dimension(sep-vert,sep-hor)

Nunca cambia el tamaño del área rígida

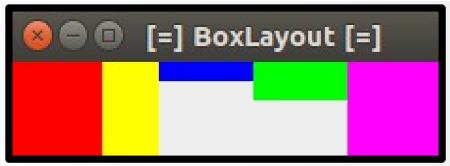
BoxLayout-Pegamento

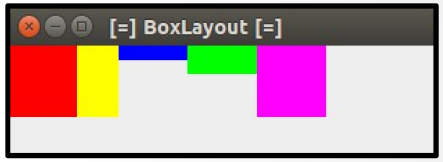
```
public void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
  mainPanel.add(createPanel(Color.red, 50, 50));

  mainPanel.add(createPanel(Color.yellow, 5, 50));
  mainPanel.add(createPanel(Color.blue, 50, 10));

  mainPanel.add(createPanel(Color.green, 50, 20));
  mainPanel.add(createPanel(Color.magenta, 50, 50));
  ...
}

private JPanel createPanel(Color color, int x, int y) {
  JPanel panel = new JPanel();
  panel.setBackground(color);
  panel.setAlignmentY(JPanel.TOP_ALIGNMENT);
  panel.setPreferredSize (new Dimension(x, y));
  panel.setMaximumSize(new Dimension(x, y));
  panel.setMinimumSize(new Dimension(x, y));
  return panel;
}
```



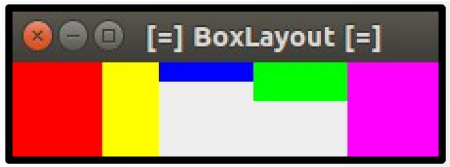


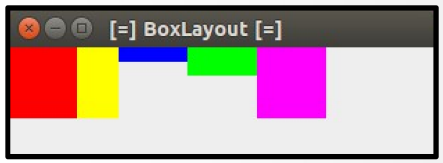
Sin pegamento

BoxLayout-Pegamento

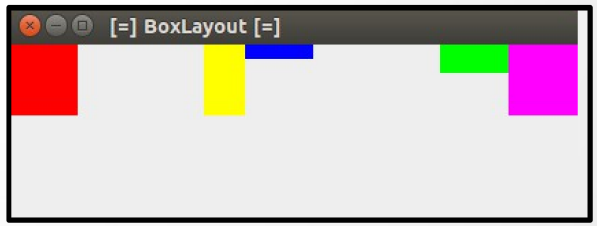
```
public void initGUI() {
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel, BoxLayout.X_AXIS));
  mainPanel.add(createPanel(Color.red, 50, 50));
  mainPanel.add(Box.createHorizontalGlue());
  mainPanel.add(createPanel(Color.yellow, 5, 50));
  mainPanel.add(createPanel(Color.blue, 50, 10));
  mainPanel.add(Box.createHorizontalGlue());
  mainPanel.add(createPanel(Color.green, 50, 20));
  mainPanel.add(createPanel(Color.magenta, 50, 50));
  ...
}

private JPanel createPanel(Color color, int x, int y) {
  JPanel panel = new JPanel();
  panel.setBackground(color);
  panel.setAlignmentY(JPanel.TOP_ALIGNMENT);
  panel.setPreferredSize (new Dimension(x, y));
  panel.setMaximumSize(new Dimension(x, y));
  panel.setMinimumSize(new Dimension(x, y));
  return panel;
}
```





Sin pegamento



Con pegamento

GridLayout

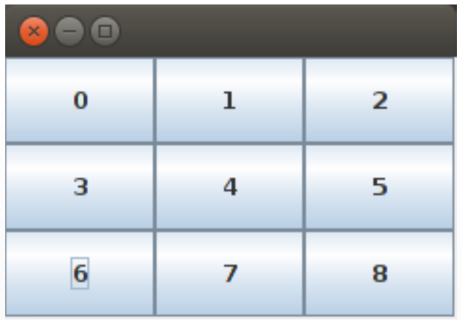
- Coloca las componentes en forma de matriz, especificando en la constructora el número de filas y columnas que se desean.

```
new GridLayout(NumFilas, NumColumnas)
new GridLayout(NumFIlas, NumColumnas, SepColumnas, SepFilas)
```

- En cada casilla solo puede haber un componente. Pero en particular puede ser un **JPanel**, que a su vez puede contener más componentes.
- Los componentes se añaden de izquierda a derecha (o de derecha a izquierda) y además de arriba a abajo.
- Al redimensionar, las componentes añadidas en cada casilla se redimensionan de acuerdo al tamaño de la ventana.



Estado normal



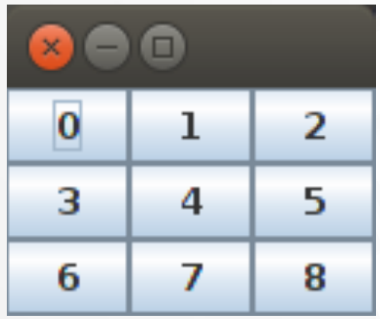
Redimensionado



Con separación horizontal y vertical

GridLayout

```
private void initGUI() {
  JFrame.setDefaultLookAndFeelDecorated(true);
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.setLayout(new GridLayout(3,3));
  for (int i=0;i<9;i++) {
    JButton button = new JButton(Integer.toString(i));
    this.add(button);
  }
  this.setVisible(true);
  this.pack();
}
```



GridLayout

```
private void initGUI() {
  JFrame.setDefaultLookAndFeelDecorated(true);
  this.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
  this.setLayout(new GridLayout(3,3,5,10));
  for (int i=0;i<9;i++) {
    JButton button = new JButton(Integer.toString(i));
    this.add(button);
  }
  this.setVisible(true);
  this.pack();
}
```

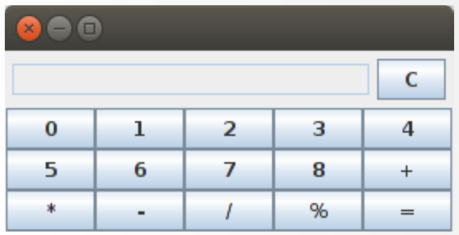


GridLayout

```
private void initGUI() {




}
```



GridLayout

```
private void initGUI() {
  ...
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel,BoxLayout.Y_AXIS));
```




```
}
```

GridLayout

```
private void initGUI() {
  ...
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel,BoxLayout.Y_AXIS));
  JPanel supPanel = new JPanel(new FlowLayout());
  textfield  = new JTextField(20);
  textfield .setEditable (false );
  JButton clear  = new JButton("C");
  clear .addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent arg0) {
      textfield .setText("");
    }
  });
  supPanel.add(textfield );
  supPanel.add(clear );




}
```



GridLayout

```
private void initGUI() {
  ...
  JPanel mainPanel = new JPanel();
  mainPanel.setLayout(new BoxLayout(mainPanel,BoxLayout.Y_AXIS));
  JPanel supPanel = new JPanel(new FlowLayout());
  textfield  = new JTextField(20);
  textfield .setEditable (false );
  JButton clear  = new JButton("C");
  clear .addActionListener(new ActionListener() {
    @Override
    public void actionPerformed(ActionEvent arg0) {
      textfield .setText("");
    }
  });
  supPanel.add(textfield );
  supPanel.add(clear );
  JPanel keyBoard = new JPanel();
  keyBoard.setLayout(new GridLayout(3,5));
  // add buttons to keyBoard
  mainPanel.add(supPanel);
  mainPanel.add(keyBoard);
  this .add(mainPanel);
  ...
}
```



GridLayout

```
    // add buttons to keyBoard

    for (int i=0;i<9;i++) {
      this.addButton(Integer.toString(i), keyBoard);
      this.addButton("+",keyBoard);
      this.addButton("*",keyBoard);
      this.addButton("-",keyBoard);
      this.addButton("/",keyBoard);
      this.addButton("%",keyBoard);
      this.addButton("=",keyBoard);
    }
}

private void addButton(String caption, JPanel panel) {
    JButton button = new JButton(caption);
    button.addActionListener(new ActionListener() {
      @Override
      public void actionPerformed(ActionEvent arg0) {
        String currentText = textfield.getText();
        textfield.setText(currentText+ caption);
      }
    });
    panel.add(button);
}
```



CardLayout

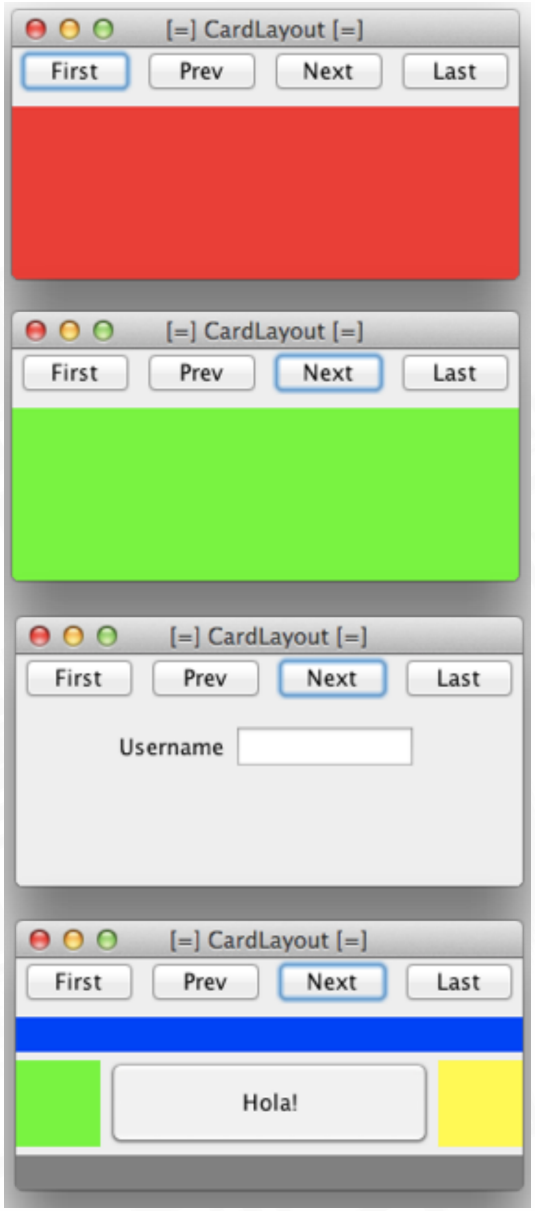
```
private void initGUI() {
  JPanel mainPanel = new JPanel(new BorderLayout());

  buttonsPanel = createButtons();
  mainPanel.add(buttonsPanel, BorderLayout.PAGE_START);

  cardsPanel = new JPanel(new CardLayout());
  mainPanel.add(cardsPanel, BorderLayout.CENTER);

  cardsPanel.add( createPanel(...)   ,"Card 1");
  ...
  cardsPanel.add( someContent() ,"Card 4");
  ...
}

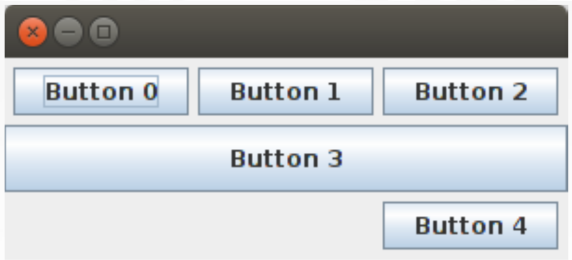
public void actionPerformed(ActionEvent e) {
  CardLayout cl = (CardLayout)(cardsPanel.getLayout());
  if ( e.getSource() == nextButton ) {
    cl.next(cardsPanel);
  } else if ( e.getSource() == prevButton ) {
    cl.previous(cardsPanel);
  } else if ( e.getSource() == firstButton ) {
    cl.first (cardsPanel);
  } else if ( e.getSource() == lastButton ) {
    cl.last (cardsPanel);
  }
}
```



Ejemplo 1

```
private void initGUI() {

}
```



Ejemplo 1

```
private void initGUI(){
  JPanel mainPanel = new JPanel(new GridLayout(3,1));

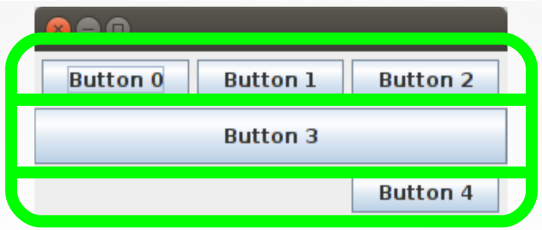








}
```



Ejemplo 1

```
private void initGUI(){
  JPanel mainPanel = new JPanel(new GridLayout(3,1));

  JPanel supPanel = new JPanel(new FlowLayout());
  supPanel.add(this.createButton("Button 0"));
  supPanel.add(this.createButton("Button 1"));
  supPanel.add(this.createButton("Button 2"));

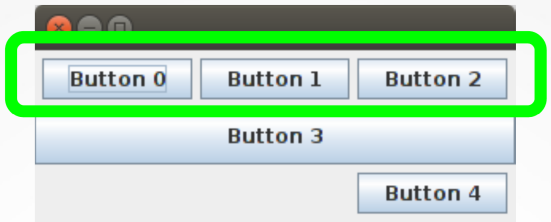








}
```



Ejemplo 1

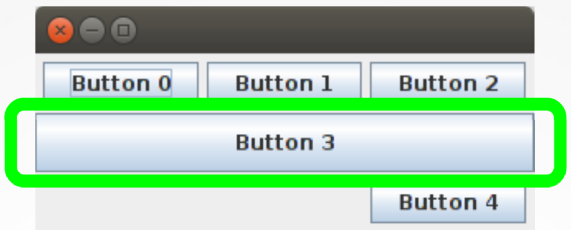
```
private void initGUI(){
  JPanel mainPanel = new JPanel(new GridLayout(3,1));

  JPanel supPanel = new JPanel(new FlowLayout());
  supPanel.add(this.createButton("Button 0"));
  supPanel.add(this.createButton("Button 1"));
  supPanel.add(this.createButton("Button 2"));

  JPanel midPanel = new JPanel(new GridLayout(1,1));
  midPanel.add(this.createButton("Button 3"));




}
```



Ejemplo 1

```
private void initGUI(){
  JPanel mainPanel = new JPanel(new GridLayout(3,1));

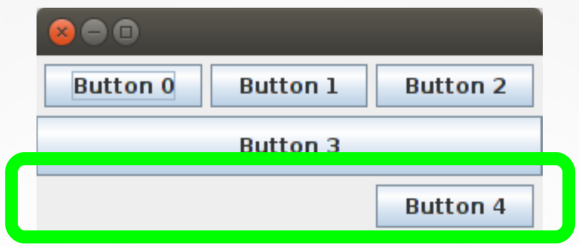
  JPanel supPanel = new JPanel(new FlowLayout());
  supPanel.add(this.createButton("Button 0"));
  supPanel.add(this.createButton("Button 1"));
  supPanel.add(this.createButton("Button 2"));

  JPanel midPanel = new JPanel(new GridLayout(1,1));
  midPanel.add(this.createButton("Button 3"));

  JPanel infPanel  = new JPanel(new FlowLayout(FlowLayout.RIGHT));
  infPanel.add(this.createButton("Button 4"));



}
```



Ejemplo 1

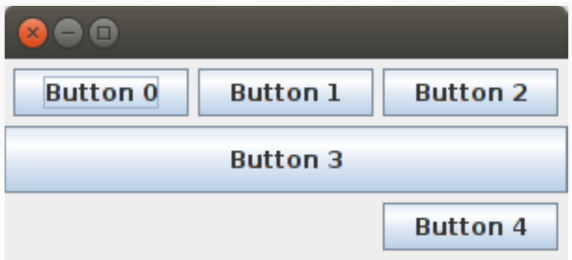
```
private void initGUI(){
  JPanel mainPanel = new JPanel(new GridLayout(3,1));

  JPanel supPanel = new JPanel(new FlowLayout());
  supPanel.add(this.createButton("Button 0"));
  supPanel.add(this.createButton("Button 1"));
  supPanel.add(this.createButton("Button 2"));

  JPanel midPanel = new JPanel(new GridLayout(1,1));
  midPanel.add(this.createButton("Button 3"));

  JPanel infPanel  = new JPanel(new FlowLayout(FlowLayout.RIGHT));
  infPanel.add(this.createButton("Button 4"));

  mainPanel.add(supPanel);
  mainPanel.add(midPanel);
  mainPanel.add(infPanel);
  this.add(mainPanel);
}
```



Ejemplo 2

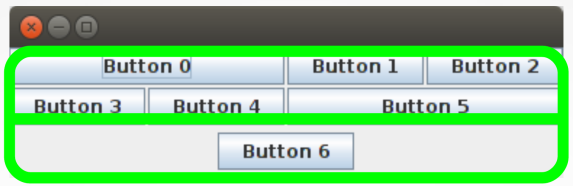
```
private void initGUI() {

}
```



Ejemplo 2

```
private void initGUI(){
  JPanel mainPanel = new JPanel();
  BoxLayout layout = new BoxLayout(mainPanel,BoxLayout.Y_AXIS);
  mainPanel.setLayout(layout);
}
```

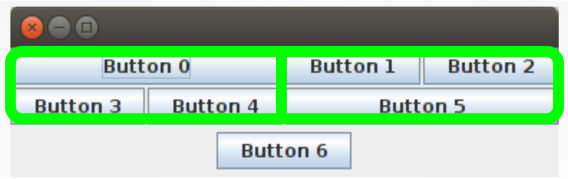


Ejemplo 2

```
private void initGUI(){
  JPanel mainPanel = new JPanel();
  BoxLayout layout = new BoxLayout(mainPanel,BoxLayout.Y_AXIS);
  mainPanel.setLayout(layout );
  JPanel supPanel = new JPanel(new GridLayout(1,2,2,2));




}
```



Ejemplo 2

```
private void initGUI(){
  JPanel mainPanel = new JPanel();
  BoxLayout layout = new BoxLayout(mainPanel,BoxLayout.Y_AXIS);
  mainPanel.setLayout(layout);
  JPanel supPanel = new JPanel(new GridLayout(1,2,2,2));
  JPanel izqPanel = new JPanel(new GridLayout(2,1,2,2));
  izqPanel.add(this.createButton("Button 0"));
  JPanel izqInfPanel = new JPanel(new GridLayout(1,2,2,2));
  izqInfPanel.add(this.createButton("Button 3"));
  izqInfPanel.add(this.createButton("Button 4"));
  izqPanel.add(izqInfPanel);
  supPanel.add(izqPanel);




}
```



Ejemplo 2

```
private void initGUI(){
  JPanel mainPanel = new JPanel();
  BoxLayout layout = new BoxLayout(mainPanel,BoxLayout.Y_AXIS);
  mainPanel.setLayout(layout);
  JPanel supPanel = new JPanel(new GridLayout(1,2,2,2));
  JPanel izqPanel = new JPanel(new GridLayout(2,1,2,2));
  izqPanel.add(this.createButton("Button 0"));
  JPanel izqInfPanel = new JPanel(new GridLayout(1,2,2,2));
  izqInfPanel.add(this.createButton("Button 3"));
  izqInfPanel.add(this.createButton("Button 4"));
  izqPanel.add(izqInfPanel);
  supPanel.add(izqPanel);
  JPanel derPanel = new JPanel(new GridLayout(2,1,2,2));
  JPanel derSupPanel = new JPanel(new GridLayout(1,2,2,2));
  derSupPanel.add(this.createButton("Button 1"));
  derSupPanel.add(this.createButton("Button 2"));
  derPanel.add(derSupPanel);
  derPanel.add(this.createButton("Button 5"));
  supPanel.add(derPanel);
  mainPanel.add(supPanel);



}
```



Ejemplo 2

```
private void initGUI(){
  JPanel mainPanel = new JPanel();
  BoxLayout layout = new BoxLayout(mainPanel,BoxLayout.Y_AXIS);
  mainPanel.setLayout(layout);
  JPanel supPanel = new JPanel(new GridLayout(1,2,2,2));
  JPanel izqPanel = new JPanel(new GridLayout(2,1,2,2));
  izqPanel.add(this.createButton("Button 0"));
  JPanel izqInfPanel = new JPanel(new GridLayout(1,2,2,2));
  izqInfPanel.add(this.createButton("Button 3"));
  izqInfPanel.add(this.createButton("Button 4"));
  izqPanel.add(izqInfPanel);
  supPanel.add(izqPanel);
  JPanel derPanel = new JPanel(new GridLayout(2,1,2,2));
  JPanel derSupPanel = new JPanel(new GridLayout(1,2,2,2));
  derSupPanel.add(this.createButton("Button 1"));
  derSupPanel.add(this.createButton("Button 2"));
  derPanel.add(derSupPanel);
  derPanel.add(this.createButton("Button 5"));
  supPanel.add(derPanel);
  mainPanel.add(supPanel);
  JPanel infPanel = new JPanel(new FlowLayout());
  infPanel.add(this.createButton("Button 6"));
  mainPanel.add(infPanel);
  this.add(mainPanel);
}
```

